

ECM Summary from Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

family=ECM

rows=8

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: ECM_Summary

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
butter	silpo_novus	observed	daily	prozorro_to_retail	ECM	retail	prozorro	49	-0.01931753309263347	-0.1130442477150289	-0.5842642909954745	0.007904731650635205	nan	nan	nan	ok	0	engle_granger_p=0.06; lag... Primary chain is strictly...	
butter	silpo_novus	promo_controlled	daily	prozorro_to_retail	ECM	retail	prozorro	49	-0.01931753309263347	-0.1130442477150289	-0.5842642909954745	0.007904731650635205	nan	nan	nan	ok	0	engle_granger_p=0.06; lag... Primary chain is strictly...	
milk	silpo	observed	daily	producer_to_prozorro	ECM	prozorro	producer	197	-0.002155003969288638	-0.3006834081052232	-0.9538372381620318	3.295228800215539e-21	nan	nan	nan	unreliable	1	engle_granger_p=1.912e-24... Primary chain is strictly...	
milk	silpo	observed	daily	prozorro_to_retail	ECM	retail	prozorro	35	0.03780420860354025	0.07959450904544814	-0.9002155090608189	0.0004638981008127622	nan	nan	nan	ok	0	engle_granger_p=9.326e-06... Primary chain is strictly...	
milk	silpo	promo_controlled	daily	producer_to_prozorro	ECM	prozorro	producer	197	-0.002155003969288638	-0.3006834081052232	-0.9538372381620318	3.295228800215539e-21	nan	nan	nan	unreliable	1	engle_granger_p=1.912e-24... Primary chain is strictly...	
milk	silpo	promo_controlled	daily	prozorro_to_retail	ECM	retail	prozorro	35	0.03780420860354025	0.07959450904544814	-0.9002155090608189	0.0004638981008127622	nan	nan	nan	ok	0	engle_granger_p=9.326e-06... Primary chain is strictly...	
milk	novus	observed	daily	producer_to_prozorro	ECM	prozorro	producer	197	-0.002155003969288638	-0.3006834081052232	-0.9538372381620318	3.295228800215539e-21	nan	nan	nan	unreliable	1	engle_granger_p=1.912e-24... Primary chain is strictly...	
milk	novus	promo_controlled	daily	producer_to_prozorro	ECM	prozorro	producer	197	-0.002155003969288638	-0.3006834081052232	-0.9538372381620318	3.295228800215539e-21	nan	nan	nan	unreliable	1	engle_granger_p=1.912e-24... Primary chain is strictly...	